Tran The Duong

Computer Vision | Multi-Object Tracking | Robotics and Control
Daejeon, South Korea | duongtranthe1@kaist.ac.kr
github.com/duongtranth | sites.google.com/view/theduongtran

M.S. student in Electrical Engineering at KAIST with experience in robotic dynamics, optimal and stochastic control, and deep learning for multi-object tracking. Develops simulation and perception software in Python, C++, and MATLAB, from control algorithms and state estimation to Transformer-based tracking and reproducible evaluation.

Education

KAIST
M.S. in Electrical Engineering, Daejeon, South Korea Sep. 2024 – Present
Expected graduation: February 2027. Advisor: Prof. Dong Eui Chang.

Hanoi University of Science and Technology
B.S. in Automatic Control, Hanoi, Vietnam Sep. 2019 – May 2024
GPA: **3.89/4.00**. Thesis advisor: Assoc. Prof. Dao Phuong Nam.

Research Experience

Graduate Researcher, Control Laboratory, KAIST
Multi-object tracking and aerial perception Oct. 2025 – Present

- Develop Transformer-based trackers for persistent object identity in crowded scenes and drone-and-bird surveillance.
- Build dataset preparation, GPU training, tracking evaluation, and diagnostic pipelines; analyze occlusion, missed detections, identity switches, and target re-entry.

Graduate Researcher, Control Laboratory, KAIST
Robotics, optimal control, and state estimation Sep. 2024 – Oct. 2025

- Developed wheeled-bipedal robot models and control software, constrained trajectory optimization, stochastic control algorithms, and simulation tools.
- Worked with MATLAB/Simulink, Gazebo, CasADi, Python, C++, and ROS 2/PX4.

Master's Thesis

End-to-End Transformer-Based Multi-Object Tracking for Drone and Bird Surveillance
KAIST, ongoing *Oct. 2025 – Present*

- Developed **BeliefCO**, combining learned motion uncertainty and object existence with end-to-end Transformer tracking and probabilistic data association.
- Built drone-and-bird tracking annotations and recording-level splits for WOSDETC; audited shared footage and benchmark split integrity.
- Evaluated on BFT, WOSDETC, BSB, and SMOT4SB with controlled detector settings and component ablations. Test HOTA: **72.32 on BFT**, **62.91 on WOSDETC**, and **65.77 on BSB**.

Manuscripts in Preparation

Planned submission to CVPR 2027.

DUET: Overlap-Aware Likelihood-Ratio Association for Online Multi-Object Tracking
Research on maintaining object identities during overlaps in online multi-object tracking.

Lifespan: Memory-Guided Association and Track Management for Online Multi-Object Tracking
Research on identity continuity when objects disappear and subsequently return.

Robotics and Control Projects

Wheeled-Bipedal Robot Modeling and CLF-MPC Control
MATLAB/Simulink, CasADi, C++, Gazebo Sep. 2024 – Oct. 2025

- Derived and implemented nonlinear wheeled inverted pendulum dynamics, equilibrium analysis, and numerical state propagation for a wheeled-bipedal robot.
- Implemented constrained model predictive control with Control Lyapunov Function conditions, forward-velocity references, actuator limits, and control smoothness penalties.
- Generated polynomial height trajectories with prescribed endpoint conditions and analyzed position, velocity, acceleration, and jerk.
- Built a Gazebo–Simulink co-simulation interface using a C++ plugin and Protocol Buffers to exchange joint states, IMU measurements, and motor-torque commands.

RS-JiLQG for Nonlinear Stochastic Systems with Slip Jumps
Graduate stochastic control project; Python, NumPy Sep. 2024 – Oct. 2025

- Co-developed a risk-sensitive jump iterative Linear Quadratic Gaussian formulation for nonlinear systems subject to diffusion noise and intermittent slip disturbances.
- Implemented a dedicated solver with backward policy updates, feedback and feedforward control, regularization, forward rollout, and line search.
- Built wheeled inverted pendulum dynamics, closed-loop Monte Carlo simulations, risk-parameter sweeps, and result exports for expected and entropic cost.
- Added LQR baselines, extended and unscented Kalman filters, disturbance scenarios, and tools for failure-rate and conditional-value-at-risk evaluation.

PX4 Offboard Control through ROS 2
Python, rclpy, PX4 message interfaces Nov. 2024

- Implemented a ROS 2 node that publishes offboard-mode heartbeats, vehicle commands, and position setpoints while monitoring vehicle state and command acknowledgments.
- Added arming, mode-switch, takeoff/setpoint, and landing logic with configured publisher/subscriber quality of service.

Earlier Engineering and Research

Engineering Intern, Viettel Aerospace Institute
Embedded systems software Sep. 2023 – Feb. 2024

- Designed user interfaces for embedded systems and practiced engineering implementation with C++, MATLAB, and Simulink.

Undergraduate Research, HUST
Reinforcement learning and control laboratories 2022 – 2024

- **Undergraduate thesis:** studied output-feedback adaptive dynamic programming and robust integral-of-sign-error reinforcement learning control for a two-wheeled inverted pendulum.
- Investigated ADP-based trajectory tracking for wheeled mobile robots experiencing wheel slip, and wind-turbine control combining ADP with disturbance observers.
- Designed a PID controller for a resistance furnace as a laboratory project.

Additional Tracking Research

QueryTrack
Python, PyTorch, DETR, TrackLab Oct. 2025 – Present

- Developed an experimental joint detection-and-association pipeline based on the CAMELTrack/TrackLab stack, with detection-query and bounding-box cues.
- Implemented temporal aggregation, query-to-track alignment, and embedding-visualization tools for SportsMOT experiments.

Bayesian Motion Prompting
Python, PyTorch, Transformer-based tracking Oct. 2025 – Present

- Developed a motion-aware extension to MOTIP that estimates predictive trajectory uncertainty and incorporates motion compatibility into identity association.
- Built training and evaluation configurations for DanceTrack, SportsMOT, and bird flock tracking.

SieveTrack
Python, PyTorch, Deformable DETR Oct. 2025 – Present

- Developed an experimental persistent-query tracker to study identity continuity through crossings and overlaps.
- Implemented query-management variants, training objectives, and failure-analysis tools; investigated duplicate tracks, missing targets, and detection–association interactions.

Coursework and Software Projects

English Tutor for TOEIC and IELTS
React, Vite, Node.js/Express, SQLite Sep. 2026

- Built a web application with JWT authentication, vocabulary review, grammar/reading/listening practice, and progress tracking.
- Implemented a personal test library with question-paper, answer-key, audio, and image uploads; added text-based PDF parsing and editable import previews.

C++ Programming Exercises
C++, Git Apr. 2025

- Maintained a collection of introductory C++ examples for language practice and version-control workflows.

Pattern Recognition with RBM Features and Neural Classification
Python, NumPy, TensorFlow; KAIST coursework Dec. 2024

- Implemented a restricted Boltzmann machine feature-learning pipeline and neural classifier for MNIST, including sampling, data normalization, and train/validation/test splits.

Technical Skills

Programming: Python, C/C++, MATLAB/Simulink, JavaScript; familiarity with C# and Julia.

Vision and learning: PyTorch, Transformer-based detection/tracking, TensorFlow, data association, TrackEval, ablation studies.

Control and estimation: Nonlinear dynamics, MPC, CLF-based control, risk-sensitive optimal control, iLQG, LQR, ADP, reinforcement learning, EKF/UKF.

Robotics and tools: Gazebo, ROS 2/PX4, CasADi, Protocol Buffers, Git, Linux, CUDA GPU training, SLURM.

Web development: React, Express, SQLite, JWT authentication.

Languages: Vietnamese (native), English.

Awards and Honors

- Best Graduation Thesis award within the HUST thesis defense council, February 2024.
- HUST academic excellence scholarships for six semesters, 2020–2023.
- Corporate scholarship for outstanding students, 2023.